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* Array<integer><String> implementation insertion travsing

```
import java.util.Scanner;
class implement_Array{
    static Scanner sc=new Scanner(System.in);

    static int ArrayInterger[]=new int[5];
    static String ArrayString[]=new String[5];

 void enterName(){
    System.out.println("Enter Student data");
    for(int i=0;i<ArrayInterger.length;i++){
        System.out.println("enter name "+(1+i
        ));
        ArrayString[i]=sc.next();
        System.out.println("enter marks "+ArrayString[i]);
        ArrayInterger[i]=sc.nextInt();
    }}
void Print(){
    System.out.println("Students Name and Marks");
    for(int i=0;i<ArrayString.length;i++){
        System.out.println("{"+(i+1)+" Name :"+ArrayString[i]+"  Marks:
"+ArrayInterger[i]+"}");
    }
}

 public static void main(String[] args){
  implement_Array obj=new implement_Array();
  obj.enterName();
  obj.Print();
 }
}
/*
```

OUTPUT

```
Enter Student data
enter name 1
Akhtar
enter marks Akhtar
300
enter name 2
mudassir
enter marks mudassir
400
enter name 3
```

```
uzair
enter marks uzair
350
enter name 4
hassan
enter marks hassan
400
enter name 5
noor
enter marks noor
399
Students Name and Marks
{1 Name :Akhtar  Marks: 300}
{2 Name :mudassir  Marks: 400}
{3 Name :uzair  Marks: 350}
{4 Name :hassan  Marks: 400}
{5 Name :noor  Marks: 399}
*/
```

CREATE DATABASE BY ARRAY AND ALSO SEARCH FOR SPECIFIC STUDENT BY ,GPA,NAME,ID,OR SEMESTERS

```
import java.util.Scanner;
public class Students {

static Scanner sc=new Scanner(System.in);

static int size=sc.nextInt();
static String name[]=new String[size];
static String Id[]=new String[size];
static int DOBY[]=new int[size];
static float GPA[]=new float[size];
static int semester[]=new int[size];

   void enterData(){
    for(int i=0;i<size;i++){
        System.out.println("enter name:");
        name[i]=sc.next();
        System.out.println("enter id");
        Id[i]=sc.next();
        System.out.println("enter DOBY");
        DOBY[i]=sc.nextInt();
        System.out.println("enter GPA");
```

```
        GPA[i]=sc.nextFloat();
        System.out.println("enter semester
");          semester[i]=sc.nextInt();
        }
    }
 void checkGPA(){
    System.out.println("Student name and gpa with order of gpa");
    for(int i=0;i<size;i++){
        if(GPA[i]<3.00){
        System.out.println("\n less then 3.00 GPA");
            System.out.println("Name:"+name[i]+" ID :"+Id[i]+" Date of
brith:  "+DOBY[i]+" GPA : "+GPA[i]);
            }
        else if(GPA[i]>=3.00){
    System.out.println("\n more then 3.00 GPA");
            System.out.println("Name:"+name[i]+" ID :"+Id[i]+" Date of
brith:  "+DOBY[i]+" GPA : "+GPA[i]);
            }
                }
    }

 void searchStudent(){
    System.out.println("enter number for students searching in list \n enter 1
for Name \n enter 2 for semester \n enter 3 for Id\n enter 4 for DOB by year");
    int choice=sc.nextInt();
    switch(choice){

        case 1:
         System.out.print("enter  name:");
         String Name=sc.next();
              for(int i=0;i<size;i++)
              {
                if(Name.equals(name[i])){
                        System.out.println("Name:"+name[i]+" ID :"+Id[i]+"
Date of brith:  "+DOBY[i]+" GPA : "+GPA[i]);
                }
                }
             System.out.println("not found!");
            break;


        case 2:
          System.out.print("enter  semester:");
          int Semester=sc.nextInt();
                    for(int i=0;i<size;i++)
              {
                if(Semester==semester[i]){
```

```
                        System.out.println("Name:"+name[i]+" ID :"+Id[i]+"
Date of brith:  "+DOBY[i]+" GPA : "+GPA[i]);
                    }
                }
    System.out.println("not found!");
            break;


         case 3:
              System.out.print("enter  ID:");
          String ID=sc.next();
                      for(int i=0;i<size;i++)
              {
                if(ID.equals(Id[i])){
                        System.out.println("Name:"+name[i]+" ID :"+Id[i]+"
Date of brith:  "+DOBY[i]+" GPA : "+GPA[i]);
                    }
                }
            System.out.println("not found!");
            break;


       case 4:
       System.out.print("enter  name:");
     int DOB=sc.nextInt();
             for(int i=0;i<size;i++)
              {
                if(DOB==DOBY[i]){
                        System.out.println("Name:"+name[i]+" ID :"+Id[i]+"
Date of brith:  "+DOBY[i]+" GPA : "+GPA[i]);
                    }
                }
        System.out.println("not found!");
            break;
        }
    }
    public static void main(String[] args) {

            Students obj=new Students();

        obj.enterData();
        System.out.println("enter 1 for searching \n enter 2 for check GPA");

            int choice=sc.nextInt();
        switch(choice){
           case 1:
           obj.searchStudent();
           break;
```

```
           case 2:
            obj.checkGPA();
            break;
            }

    }
}
/*
   OUTPUT
3
enter name:
akhtar
enter id
3322
enter DOBY
2002
enter GPA
3.2
enter semester
3
enter name:
hassan
enter id
4422
enter DOBY
2004
enter GPA
3.8
enter semester
3
enter name:
Ahmad
enter id
4455
enter DOBY
2001
enter GPA
3.9
enter semester
3
enter 1 for searching
 enter 2 for check GPA
1
enter number for students searching in list
 enter 1 for Name
 enter 2 for semester
 enter 3 for Id
 enter 4 for DOB by year
```

```
3
enter  ID:4422
Name:hassan ID :4422 Date of brith:  2004 GPA : 3.8
 */
```

ARRAY ADD /REMOVE AND PRINT BY CHOICES

```
import java.util.Scanner;
public class arrayRemoveElement {
static  Scanner sc=new Scanner(System.in);
    static int  sizeArray=3;
    static String array[]=new String[sizeArray];

    void addElement(String data){
        for(int i=0;i<array.length;i++){
            if(array[i]==null){
                array[i]=data;
                break;
                }
            }
            }
    void printElement(){
        for(int i=0;i<array.length;i++){
            System.out.println(array[i]);

            }
        }
    void removeElement(String Remove){
        for(int i=0;i<array.length;i++){
            if(array[i]==Remove){
                for(int k=i;k<array.length-1;k++){
                    if(k==array.length-1){
                        array[k]=null;
                        }else{
                        array[k]=array[k+1];
                            }

                    }
                }
            }
            }
    void repeat(){
        System.out.println("enter for add (1) and for remove (2) and for print
(3)");
        int choice=sc.nextInt();
        if (choice==1){
       System.out.println("enter element");
```

```
        String data=sc.next();
        addElement(data);
             }
         else if(choice==2){
         System.out.println("enter element");
        String data=sc.next();
        removeElement(data);
             }else{
                 printElement();
                 }
         repeat();

         }

    public static void main(String[] args) {
         Scanner sc=new Scanner(System.in);
         System.out.println("enter array size");
         sizeArray=sc.nextInt();
   arrayRemoveElement obj=new arrayRemoveElement();
   obj.repeat();

         }


}

/*
```

OUTPUT

```
enter array size
5
enter for add (1) and for remove (2) and for print (3)
1
enter element
akhtar
enter for add (1) and for remove (2) and for print (3)
1
enter element
mudassir
enter for add (1) and for remove (2) and for print (3)
1
enter element
noor
enter for add (1) and for remove (2) and for print (3)
1
enter element
hassan
enter for add (1) and for remove (2) and for print (3)
```

```
2
enter element
noor
enter for add (1) and for remove (2) and for print (3)
3
akhtar
mudassir
noor
enter for add (1) and for remove (2) and for print (3)
 */
```

ArrayLIst implantation with the help of library

```
import java.util.Scanner;
import java.util.ArrayList;
class ArrayList_implement{
    static Scanner sc=new Scanner(System.in);
    static ArrayList<Integer> list=new ArrayList<>();
void enterData(){
    System.out.println("enter data into Array list\n enter loop time");
    int size=sc.nextInt();
   for(int i=0;i<size;i++){
    System.out.println("enter element into Arraylist");
    int element=sc.nextInt();
    list.add(element);
   }
}
void Replace_Element(){
    System.out.println("Enter element in specific index nomber");
    int indexNumber=sc.nextInt();
    System.out.println("enter insort element");
    int replaceElement=sc.nextInt();
 list.add(indexNumber,replaceElement);
}
void Print(){
    System.out.println("ArrayList");
    for(int i=0;i<list.size();i++){
        System.out.print(list.get(i)+",");
    }
    System.err.println("");
}
public static void main(String[] args){
    ArrayList_implement obj=new ArrayList_implement();
    obj.enterData();
    obj.Print();
```

```
    obj.Replace_Element();
    obj.Print();
}
}
```

/*OUTPUT:

```
 * enter data into Array list
 enter loop time
5
enter element into Arraylist
150
enter element into Arraylist
250
enter element into Arraylist
300
enter element into Arraylist
350
enter element into Arraylist
400
ArrayList
150,250,300,350,400,
Enter element in specific index nomber
4
enter insort element
000
ArrayList
150,250,300,350,0,400,
 *
 */
```

**Implementation of LinkedList without library

```
import java.util.Scanner;
class Linkedlist{
    static Scanner sc=new Scanner(System.in);
    Node head;

 class Node{
    int data;
    Node next;
    Node(int data){
        this.data=data;
        this.next=null;
    }
 }
void addlast(int data){
```

```
    Node newNode=new Node(data);
    if(head==null){
        head=newNode;
    }else{
        Node current=head;
        while(current.next!=null){
            current=current.next;
        }
        current.next=newNode;
    }
}
//add element before the node
void add(int data){
    Node newNode=new Node(data);
    if(head==null){
        head=newNode;
    }else{
        Node temp=head;
        head=newNode;
        newNode.next=temp;
    }
}
//add in the last of the nodes
void printlist(){
    Node current=head;
    while(current!=null){
        System.out.print(current.data+",");
        current=current.next;
    }
    System.err.println("");
}
//add element whre you want
void middle(int data){
    Node newNode=new Node(data);
    System.err.println("before eleemet whre you want insort enter element ");
    int element=sc.nextInt();
    Node currNode=head;
    while(currNode.data!=element){
        currNode=currNode.next;
    }
    Node currNodeN=head;
    while(currNodeN.data!=element){
        currNodeN=currNodeN.next;
    }
    currNodeN=currNodeN.next;
    currNode.next=newNode;
    newNode.next=currNodeN;
}
```

```
void Delete(){
    Node current=head;
    System.err.println("enter element before the element");
    int element=sc.nextInt();
    while(current.data!=element){
        current=current.next;
    }
    current.next=current.next.next;
}
public static void main(String[] args){
    Linkedlist o=new Linkedlist();
    o.addlast(45);
    o.add(2000);
    o.addlast(100);
    o.add(400);
    o.add(300);
   o.printlist();
    o.middle(1234);
    o.printlist();
   o.Delete();
    o.printlist();
}
}
```

{{{:OUTPUT:}}}

```
 * 300,400,2000,45,100,
before eleemet whre you want insort enter element
400
300,400,1234,2000,45,100,
enter element before the element
2000
300,400,1234,2000,100,
 *
 */
```

Implementation Linked list with help of library

```
import java.util.Scanner;
import java.util.LinkedList;
class Linkedlist{
    static Scanner sc=new Scanner(System.in);
    static LinkedList<String> linked=new LinkedList<>();

void addElement(){
    System.out.println("enter element int linked list\n enter size of loop");
    int size=sc.nextInt();
```

```
    for(int i=0;i<size;i++){
    System.out.println("enter element no:"+i);
    String element=sc.next();
    linked.add(element);
}}
void removeElement(){
    System.out.println("Enter  element to remove");
    String element=sc.next();
    linked.remove(element);
}
public static void main(String[] args) {
    Linkedlist linke=new Linkedlist();
    linke.addElement();
    System.out.println(linked);
    linke.removeElement();
    System.out.println(linked);

}
}
/*
```

OUTPUT

```
 * enter element int linked list
 enter size of loop
5
enter element no:0
pakistan
enter element no:1
afghanistan
enter element no:2
turky
enter element no:3
iran
enter element no:4
saudArbia
[pakistan, afghanistan, turky, iran, saudArbia]
Enter  element to remove
iran
[pakistan, afghanistan, turky, saudArbia]
 *
 */
```

Create database for student detail with the help of linkedlist add/remove by name,id,age or gpa and also print all data

```java
import java.util.Scanner;
import java.util.LinkedList;
public class list {
    static LinkedList<String> name=new LinkedList<>();
    static LinkedList<String> id=new LinkedList<>();
    static LinkedList<Integer> age=new LinkedList<>();
    static LinkedList<Float> gpa=new LinkedList<>();
        static LinkedList<String> semester=new LinkedList<>();
        static Scanner sc=new Scanner(System.in);
        public void enter_detail(){
            System.out.print("enter name:");
        String Name=sc.next();
                name.add(Name);

                System.out.print("enter id:");
        String ID=sc.next();
                id.add(Name);

                    System.out.print("enter age:");
        int Age=sc.nextInt();
                age.add(Age);

                System.out.print("enter gpa:");
        float GPA=sc.nextFloat();
                gpa.add(GPA);

        choice();
            }

    public void printAllData(){
            for(int i=0;i<gpa.size();i++){
                System.out.println("Name:"+name.get(i)+" id:"+id.get(i)+"
age:"+age.get(i)+" GPA:"+gpa.get(i));

                }
                    choice();
            }
    public void remove(){
        System.out.println("name press 1, id press 2, age press 3, gpa press
4");
        int choice=sc.nextInt();
        switch(choice){
        case 1:
            System.out.println("enter name:");
              String Name=sc.next();
              for(int i=0;i<=name.size();i++){
                if(name.get(i). equals(Name)){
                name.remove(i);
```

```
                age.remove(i);
                id.remove(i);
                gpa.remove(i);
                    }else if(i==name.size()){
                        System.out.println("not found");
                        choice();
                        }

                }

            }
            choice();
        }
    public void choice(){
        System.out.println("**add new data press 1 **\n **print all data press
2**\n**remove data press 3**\n");
        int CHOICE=sc.nextInt();
        switch(CHOICE){
        case 1:
             enter_detail();
             break;
        case 2:
             printAllData();
             break;
            case 3:
            remove();
            break;
            }
        }
    public static void main(String[] args) {
    list list=new list();
        list.choice();

    }
}
/*
```

OUTPUT

```
 **add new data press 1 **
 **print all data press 2**
**remove data press 3**

1
enter name:Akhtar
enter id:3345
enter age:21
enter gpa:3.2
**add new data press 1 **
```

```
 **print all data press 2**
**remove data press 3**

2
Name:Akhtar id:Akhtar age:21 GPA:3.2
**add new data press 1 **
 **print all data press 2**
**remove data press 3**

1
enter name:mudassir
enter id:33998
enter age:18
enter gpa:3.8
**add new data press 1 **
 **print all data press 2**
**remove data press 3**

3
name press 1, id press 2, age press 3, gpa press 4
1
enter name:
mudassir
 */
```

Implementation of Queue without library

```
import java.util.Scanner;
class queue{
    static int arr[]=new int[5];
    static int rear=-1;
    static int front=0;
 void enqueue(int data){
    if(rear==arr.length-1){
        System.out.println("queue is already full");
        return;
    }
    rear++;
    arr[rear]=data;
 }
 void dequeue(){
    if(front>rear){
        System.out.println("the queue is empty");
        rear=-1;
        front=0;
    }
```

```
    else{

        System.out.println("the element is dequeue::"+arr[front]);
        front++;
    }
 }
 void Print(){
    if( rear==-1){
        System.out.println("the queue is empty");
    }else{
        System.out.print("{");
    for(int i=front;i<=rear;i++){
        System.out.print(arr[i]+",");
    }
    System.out.println("}");
        }
 }
public static void main(String[] arg){
    queue o=new queue();

    o.enqueue(23);
    o.enqueue(56);
   o.Print();
    o.dequeue();
    o.dequeue();
    o.dequeue();
    o.enqueue(342);
    o.enqueue(4354);
    o.enqueue(342);
    o.enqueue(2332);
    o.enqueue(342);
    o.enqueue(565);
    o.enqueue(342);
    o.enqueue(345345);
    o.Print();
    o.dequeue();
    o.dequeue();
    o.dequeue();
    o.dequeue();
    o.dequeue();
    o.dequeue();
    o.dequeue();
    o.dequeue();
}
}
/*
```

OUTPUT

```
    {23,56,}
```

```
the element is dequeue::23
the element is dequeue::56
the queue is empty
queue is already full
queue is already full
queue is already full
{342,4354,342,2332,342,}
the element is dequeue::342
the element is dequeue::4354
the element is dequeue::342
the element is dequeue::2332
the element is dequeue::342
the queue is empty
the queue is empty
the queue is empty
 */
```

Implementation of Queue with help of library

```
_import java.util.Scanner;

import java.util.Queue;
import java.util.LinkedList;
class QUEUES{
    static Queue<Integer> queue=new LinkedList<>();
    static Scanner sc=new Scanner(System.in);

public static void main(String[] args){
    System.out.println("stack is empty :"+queue.isEmpty());
    //add element in queue
    queue.add(34);
    queue.add(29929);
    queue.add(232);
    //remove form queues
    queue.remove();
    queue.add(264234);
    //print 1st//peek vlaue
    System.out.println(queue.peek());
    System.out.println("queue is empty:"+queue.isEmpty());
    //print all queue
    System.out.println(queue);
}
}
/*
```

OUTPUT

```
 * stack is empty :true
```

```
    29929
    queue is empty:false
    [29929, 232, 264234]
 */
```

Implementation of priority Queue without library

```
import java.util.Scanner;
class Priority_Queue{
    static int array[]=new int[5];
    static Scanner sc=new Scanner(System.in);
    int rear=-1;
    int front=0;;

void enqueue(){
    if(rear<array.length-1){
        rear++;
        System.out.println("enter numbers");
        array[rear]=sc.nextInt();
        for(int i=rear;i>front;i--){
            if(array[i]>array[i-1]){
                int temp=array[i-1];
                array[i-1]=array[i];
                array[i]=temp;
                System.out.print("{");
                for(int j=front;j<=rear;j++){
                   System.out.print(array[j]+",");
                }
                System.out.print("}\n\n\n");
            }
        }

    }
    else{
        System.out.println("the array is full"+rear);
    }
}
void dequeue(){
    if(rear==-1){
        System.out.println("the queue is empty");
    }
    else{
       System.out.println(front+"the "+array[front]+" is delete");
       front++;
    }
    if(front>rear){
```

```
        rear=-1;
        front=0;
    }

}
void peek(){
    System.out.println(array[front]);
}
void print(){
    System.out.print(" {}{}{{{{{{{print}}}}}}}}\n{");
    for( int i=front;i<=rear;i++){
        System.out.print(array[i]+" ,");
    }
    System.out.println("}");
}
public static void main(String[] arg){
    Priority_Queue obj=new Priority_Queue();
    obj.enqueue();
    obj.enqueue();
    obj.enqueue();
    obj.enqueue();
    obj.enqueue();
    obj.enqueue();
    obj.enqueue();
    obj.print();
    obj.dequeue();
    obj.peek();
    obj.dequeue();
    obj.dequeue();
    obj.dequeue();
    obj.dequeue();
    obj.dequeue();
    obj.dequeue();
    obj.dequeue();
    System.out.println("front:"+obj.front+" and rear= "+obj.rear);
    obj.enqueue();
    System.out.println("front:"+obj.front+" and rear= "+obj.rear);
}
}
/*
```

OUTPUT

```
    enter numbers
324
enter numbers
345
{345,324,}
```

```
enter numbers
123
enter numbers
6
enter numbers
234
{345,324,123,234,6,}


{345,324,234,123,6,}


the array is full4
the array is full4
 {}{}{{{{{{print}}}}}}}}
{345 ,324 ,234 ,123 ,6 ,}
0the 345 is delete
324
1the 324 is delete
2the 234 is delete
3the 123 is delete
4the 6 is delete
the queue is empty
the queue is empty
the queue is empty
front:0 and rear= -1
enter numbers
6657
front:0 and rear= 0

 */
```

ADD data remove data from queue by chioces

```
import java.util.Scanner;
public class Queues{
      public static class queue {

           static String arr[];
         static  int size;
          static int fear;
         static int front;
          queue(int size){
            this.size=size;
            arr=new String[size];
            fear=-1;
```

```
            front=0;
            }
     public static boolean isEmpty(){
            return fear==-1;
         }
      public static void add(String data){
            if(fear==size-1){
                System.out.println("the queue is alredy full");
                return ;
                }
            fear++;
       arr[fear]=data;

            }
    public static void remove(){
            if(isEmpty()){
            System.out.println("the queue is alredy empty");
            return;
                }
            arr[front]=null;
            for(int i=0;i<fear;i++){
                arr[i]=arr[i+1];
                }
            arr[fear]=null;
            fear--;
            }
        public static String peek(){
              if(isEmpty()){
                String re="The queue is empty ";
              return re;
                }
              return arr[front];

                }
                }
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        queue queues=new queue(5);
        for(int i=0;i<100;i++){
    System.out.println("add(1),,  remove(2),,  check is
empty(3),,  peek(4).                      ,,peekAllData(5)");
int choice=sc.nextInt();
switch(choice){
    case 1:
        System.out.println("enter data");
        String data=sc.next();
        queues.add(data);
        break;
```

```
    case 2:
        System.out.println("frist element is remove");
        queues.remove();
        break;

    case 3:
       if(queues.isEmpty()){
        System.out.println("yes the queue is now empty ");
        return ;
        }
       System.out.println("No in queue are "+(queues.fear+1)+"elements");
        break;
   case 4:
    System.out.println(queues.peek());
    break;
   case 5:
   int duplicate=queues.front;
    while(duplicate<=queues.fear){
        System.out.println(queues.arr[duplicate]);
        duplicate++;
        }
}
}
    }
}
/*
     OUTPUT
a
add(1),,  remove(2),,  check is
empty(3),,  peek(4).                        ,,peekAllData(5)
1
enter data
Akhtar
add(1),,  remove(2),,  check is
empty(3),,  peek(4).                        ,,peekAllData(5)
1
enter data
Mudassir
add(1),,  remove(2),,  check is
empty(3),,  peek(4).                        ,,peekAllData(5)
1
enter data
ahmad
add(1),,  remove(2),,  check is
empty(3),,  peek(4).                        ,,peekAllData(5)
1
enter data
```

```
Hassan
add(1),,  remove(2),,  check is
empty(3),,  peek(4).                          ,,peekAllData(5)
2
frist element is remove
add(1),,  remove(2),,  check is
empty(3),,  peek(4).                          ,,peekAllData(5)
3
No in queue are 3elements
add(1),,  remove(2),,  check is
empty(3),,  peek(4).                          ,,peekAllData(5)
4
Mudassir
add(1),,  remove(2),,  check is
empty(3),,  peek(4).                          ,,peekAllData(5)
5
Mudassir
ahmad
Hassan
add(1),,  remove(2),,  check is
empty(3),,  peek(4).                          ,,peekAllData(5)
 */
```

Implementation of Stack without library

```
import java.util.Scanner;
class Priority_Queue{
    static int array[]=new int[5];
    static Scanner sc=new Scanner(System.in);
    int rear=-1;
    int front=0;;

void enqueue(){
    if(rear<array.length-1){
        rear++;
        System.out.println("enter numbers");
        array[rear]=sc.nextInt();
        for(int i=rear;i>front;i--){
            if(array[i]>array[i-1]){
                int temp=array[i-1];
                array[i-1]=array[i];
                array[i]=temp;
                System.out.print("{");
                for(int j=front;j<=rear;j++){
```

```
                    System.out.print(array[j]+",");
                }
                System.out.print("}\n\n\n");
            }
        }

    }
    else{
        System.out.println("the array is full"+rear);
    }
}
void dequeue(){
    if(rear==-1){
        System.out.println("the queue is empty");
    }
    else{
       System.out.println(front+"the "+array[front]+" is delete");
       front++;
    }
    if(front>rear){
        rear=-1;
        front=0;
    }

}
void peek(){
    System.out.println(array[front]);
}
void print(){
    System.out.print(" {}{}{{{{{{{print}}}}}}}}\n{");
    for( int i=front;i<=rear;i++){
        System.out.print(array[i]+" ,");
    }
    System.out.println("}");
}
public static void main(String[] arg){
    Priority_Queue obj=new Priority_Queue();
    obj.enqueue();
    obj.enqueue();
    obj.enqueue();
    obj.enqueue();
    obj.enqueue();
    obj.enqueue();
    obj.enqueue();
    obj.print();
    obj.dequeue();
    obj.peek();
    obj.dequeue();
```

```
    obj.dequeue();
    obj.dequeue();
    obj.dequeue();
    obj.dequeue();
    obj.dequeue();
    obj.dequeue();
    System.out.println("front:"+obj.front+" and rear= "+obj.rear);
    obj.enqueue();
    System.out.println("front:"+obj.front+" and rear= "+obj.rear);
}
}
/*
    enter numbers
324
enter numbers
345
{345,324,}


enter numbers
123
enter numbers
6
enter numbers
234
{345,324,123,234,6,}


{345,324,234,123,6,}


the array is full4
the array is full4
 {}{}{{{{{print}}}}}}}}
{345 ,324 ,234 ,123 ,6 ,}
0the 345 is delete
324
1the 324 is delete
2the 234 is delete
3the 123 is delete
4the 6 is delete
the queue is empty
the queue is empty
the queue is empty
front:0 and rear= -1
enter numbers
6657
front:0 and rear= 0
```

```
 */
```

Implementation of Stack with of library

```
import java.util.Scanner;
//stack libraRY
import java.util.Stack;
class STACK{
  public static void main(String[] args){
   Stack<Integer> stack=new Stack<>();
   //insort element into stack
   stack.push(2);
   stack.push(3);
   stack.push(44);
   stack.push(500);
   System.out.println(stack);
   //pop element
   stack.pop();
   //new insort in top
   stack.push(40000);
   System.out.print(stack);
    }
}
/*
 * OUTPUT:
 *        [2, 3, 44, 500]
          [2, 3, 44, 40000]
 */
```

Implementation of Stack on LinkedList

```
import java.util.Scanner;
class Linkedlist{
    static Scanner sc=new Scanner(System.in);
    Node head;

 class Node{
    int data;
    Node next;
    Node(int data){
        this.data=data;
        this.next=null;
    }
```

```
 }
void addlast(int data){
    Node newNode=new Node(data);
    if(head==null){
        head=newNode;
    }else{
        Node current=head;
        while(current.next!=null){
            current=current.next;
        }
        current.next=newNode;
    }
}
//add element before the node
void add(int data){
    Node newNode=new Node(data);
    if(head==null){
        head=newNode;
    }else{
        Node temp=head;
        head=newNode;
        newNode.next=temp;
    }
}
//add in the last of the nodes
void printlist(){
    Node current=head;
    while(current!=null){
        System.out.print(current.data+",");
        current=current.next;
    }
    System.err.println("");
}
//add element whre you want
void middle(int data){
    Node newNode=new Node(data);
    System.err.println("before eleemet whre you want insort enter element ");
    int element=sc.nextInt();
    Node currNode=head;
    while(currNode.next!=null){
        if(currNode.data==element){
        currNode=currNode.next;
        }
    }
    Node currNodeN=head;
    while(currNodeN.next!=null){
        if(currNodeN.data==element){
```

```
        }
        currNodeN=currNodeN.next;
    }
    currNodeN=currNodeN.next;
    currNode.next=newNode;
    newNode.next=currNodeN;
}
void Delete(){
    Node current=head;
    System.err.println("enter element before the element");
    int element=sc.nextInt();
    while(current.data!=element){
        current=current.next;
    }
    current.next=current.next.next;
}
public static void main(String[] args){
    Linkedlist o=new Linkedlist();
    o.addlast(45);
    o.add(2000);
    o.addlast(100);
    o.add(400);
    o.add(300);
   o.printlist();
    o.middle(1234);
    o.printlist();
   o.Delete();
    o.printlist();
}
}
/*
```

OUTPUT

```
 * 300,400,2000,45,100,
before eleemet whre you want insort enter element
400
300,400,1234,2000,45,100,
enter element before the element
2000
300,400,1234,2000,100,
 *
 */
```

SORTING

BOBBLE SORT

```
import java.util.Scanner;
class  BUBBLESORT{
    static int array[]=new int[10];
    static Scanner sc=new Scanner(System.in);
void InsortData(){
    System.out.println("enter element into Array");
    for(int i=0;i<array.length;i++){
      System.out.println("enter element no: "+i);
      array[i]=sc.nextInt();
    }
}
void Print(){
    System.out.print("Arrray: \n {");
    for(int i=0;i<array.length;i++){
        System.out.print(array[i]+",");
    }
    System.out.println("}");
}
//BUBBLE DECREASE
void BUBBLE(){
    for(int i=0;i<array.length;i++){
        for(int j=0; j<array.length-i-1;j++){
            if(array[j]>array[j+1]){
                int temp=array[j];
                array[j]=array[j+1];
                array[j+1]=temp;
            }
        }
    }
}
void BUBBLEA(){
    for(int i=0;i<array.length;i++){
        for(int j=0;j<array.length-i-1;j++){
            if(array[j]<array[j+1]){
                int temp=array[j];
                array[j]=array[j+1];
                array[j+1]=temp;
            }
        }
    }
}
public static void main(String[] args) {
    BUBBLESORT obj=new BUBBLESORT();
    obj.InsortData();
    System.out.println("orignal");
    obj.Print();
```

```
    obj.BUBBLE();
    System.out.println("ACREAMENt");
    obj.Print();
    obj.BUBBLEA();
    System.out.println("DECREAMENT");
    obj.Print();
}
}
/*
```

OUTPUT

```
 enter element into Array
enter element no: 0
345
enter element no: 1
657
enter element no: 2
234
enter element no: 3
76898
enter element no: 4
534
enter element no: 5
34
enter element no: 6
534
enter element no: 7
6976
enter element no: 8

342
enter element no: 9
65
orignal
Arrray:
 {345,657,234,76898,534,34,534,6976,342,65,}
ACREAMENt
Arrray:
 {34,65,234,342,345,534,534,657,6976,76898,}
DECREAMENT
Arrray:
 {76898,6976,657,534,534,345,342,234,65,34,}

 */
```

SELECTION SORT

```java
import java.util.Scanner;
class SLECTION{
    static int array[]=new int[10];
     static Scanner sc=new Scanner(System.in);
void insort(){
    System.out.println("insort element");
   for(int i=0;i<array.length;i++){
    System.out.println("enter element no "+i);
    array[i]=sc.nextInt();
   }
}
void slectionSort(){
    for(int i=0;i<array.length;i++){
        int temp=array[i];
        int lessindx=i;
        for(int j=i;j<array.length;j++){
            if(array[lessindx]>array[j]){
                lessindx=j;
            }
        }
        array[i]=array[lessindx];
        array[lessindx]=temp;
    }

}
void DslectionSort(){
    for(int i=0;i<array.length;i++){
        int temp=array[i];
        int graterindx=i;
        for(int j=i;j<array.length;j++){
            if(array[graterindx]<array[j]){
                graterindx=j;
            }
        }
        array[i]=array[graterindx];
        array[graterindx]=temp;
    }

}
void print(){
    for(int i=0;i<array.length;i++){
        System.out.print(array[i]+",");
    }
    System.out.println("");
}
public static void main(String[] args) {
    SLECTION obj=new SLECTION();
    System.out.println("orignal");
```

```
    obj.insort();
    obj.print();
    System.out.println("sort Acreament");
    obj.slectionSort();
    obj.print();
    System.out.println("Sort decreamenr");
    obj.DslectionSort();
    obj.print();
}
}
/*
```

OUTPUT

```
orignal
insort element
enter element no 0
34
enter element no 1
66573
enter element no 2
234
enter element no 3
5342
enter element no 4
234
enter element no 5
7654
enter element no 6
34
enter element no 7
87
enter element no 8

43
enter element no 9
78
34,66573,234,5342,234,7654,34,87,43,78,
sort Acreament
34,34,43,78,87,234,234,5342,7654,66573,
Sort decreamenr
66573,7654,5342,234,234,87,78,43,34,34,
 */
```

INSORTION SORT

```
import java.util.Scanner;
```

```
class InsortionSort{
    static int array[]=new int[5];
    static Scanner sc=new Scanner(System.in);
void Array(){
    System.out.println("enter elements");
    for(int i=0;i<array.length;i++){
        System.out.print("value"+(i+1)+":  ");
        array[i]=sc.nextInt();
    }
}
void InsortionA(){
    for(int i=1;i<array.length;i++){
    int pr=i-1;int cur=i;
    while(pr>=0 && array[pr]>=array[cur]){
        System.out.println("privous:"+(pr+1)+" current:"+(cur+1));
        int temp=array[cur];
        array[cur]=array[pr];
        array[pr]=temp;
        pr--;
        cur--;
        print();
    }
}
}
void InsortionD(){
    for(int i=1;i<array.length;i++){
        int pr=i-1;
        int curr=i;
        while(pr>=0 && array[pr]<=array[curr]){
            System.out.println("privous:"+(pr+1)+" current:"+(curr+1));
            int temp=array[curr];
            array[curr]=array[pr];
            array[pr]=temp;
            pr--;
            curr--;
            print();
        }
    }
}
void print(){
    System.out.println("array");
    for(int i=0;i<array.length;i++){
        System.out.print(array[i]+",");
    }
    System.out.println("");
}
public static void main(String[] args){
    InsortionSort o=new InsortionSort();
```

```
    o.Array();
    o.print();
    System.out.println("increasing");
    o.InsortionA();
    o.print();
    System.out.println("Decreasing");
    o.InsortionD();
    o.print();
}
}
/*
```

OUTPUT

```
 enter elements
value1:  868
value2:  98879
value3:  435
value4:  980
value5:  868
array
868,98879,435,980,868,
increasing
privous:2 current:3
array
868,435,98879,980,868,
privous:1 current:2
array
435,868,98879,980,868,
privous:3 current:4
array
435,868,980,98879,868,
privous:4 current:5
array
435,868,980,868,98879,
privous:3 current:4
array
435,868,868,980,98879,
privous:2 current:3
array
435,868,868,980,98879,
array
435,868,868,980,98879,
Decreasing
privous:1 current:2
array
868,435,868,980,98879,
privous:2 current:3
array
```

```
868,868,435,980,98879,
privous:1 current:2
array
868,868,435,980,98879,
privous:3 current:4
array
868,868,980,435,98879,
privous:2 current:3
array
868,980,868,435,98879,
privous:1 current:2
array
980,868,868,435,98879,
privous:4 current:5
array
980,868,868,98879,435,
privous:3 current:4
array
980,868,98879,868,435,
privous:2 current:3
array
980,98879,868,868,435,
privous:1 current:2
array
98879,980,868,868,435,
array
98879,980,868,868,435,
 */
```

MERGE SORT

```
class mergeSort{
    static int[] array={23,87,34,89,56};

    void Divide(int start,int end){
        if(start<end){
            int mid=start+(end-start)/2;
            Divide(start,mid);
            Divide(mid+1,end);
            Conquer(start,mid,end);
        }
    }
    void Conquer(int start,int mid,int end){
        int[] temp=new int[end-start+1];
        int i=start;
        int j=mid+1;
```

```
        int k=0;

        while(i<=mid && j<=end){
            if(array[i]>=array[j]){
                temp[k++]=array[i++];
            }else{
                temp[k++]=array[j++];
            }
        }
        while(i<=mid){
            temp[k++]=array[i++];
        }
        while(j<=end){
            temp[k++]=array[j++];
        }

    for(int g=start,u=0;g<=end;g++,u++){
        array[g]=temp[u];
    }
    }

    void DivideD(int start,int end){
        if(start<end){
            int mid=start+(end-start)/2;
            DivideD(start,mid);
            DivideD(mid+1,end);
            ConquerD(start,mid,end);
        }
    }
    void ConquerD(int start,int mid,int end){
        int[] temp=new int[end-start+1];
        int i=start;
        int j=mid+1;
        int k=0;

        while(i<=mid && j<=end){
            if(array[i]<=array[j]){
                temp[k++]=array[i++];
            }else{
                temp[k++]=array[j++];
            }
        }
        while(i<=mid){
            temp[k++]=array[i++];
        }
        while(j<=end){
            temp[k++]=array[j++];
        }
```

```
    for(int g=start,u=0;g<=end;g++,u++){
        array[g]=temp[u];
    }
    }
 void print(){
    System.out.print("{");
    for(int i=0;i<array.length;i++){
        System.out.print(array[i]+",");
    }
    System.out.println("}");
 }
 public static void main(String[] args){
    mergeSort obj=new mergeSort();
    obj.print();
    obj.Divide( 0,array.length-1);
    System.out.println("Decending order");
    obj.print();
    System.out.println("Ascending Order");
    obj.DivideD(0,array.length-1);
    obj.print();
}
}
/*
```

OUTPUT

```
 {23,87,34,89,56,}
Decending order
{89,87,56,34,23,}
Ascending Order
{23,34,56,87,89,}
 */
```

QUICKSORT

```
      class quick{

    static int[] array={23,64,23,7,8,34,1,543,36,34,34,24,23};

void quickSort(int start,int end){
    if(start<end){
        int pivot=partition(start,end);
        quickSort(start,pivot-1);
        quickSort(pivot+1,end);
```

```
    }
}
int partition(int start,int end){
    int pivot=array[end];
    int j=start-1;
    printd(start,end);
  for(int i=start;i<end;i++){
    if(array[i]<=pivot){
        j++;
        int temp=array[i];
        array[i]=array[j];
        array[j]=temp;
    }
  }
  j++;
  array[end]=array[j];
  array[j]=pivot;
  printd(start,end);
  return j;

}
void printd(int start,int end){
    System.out.print("{");
    for(int i=start;i<=end;i++){
        System.out.print(array[i]+",");
    }
    System.out.println("]");
}
void print(){
    System.out.print("{");
    for(int j=0;j<array.length;j++){
        System.out.print(array[j]+",");
    }
    System.out.println("}");
}
public static void main(String[] args){
    quick obj=new quick();
    obj.quickSort(0,array.length-1);
    obj.print();
}
}
/*
```

{{{{OUTPUT}}}}

```
{23,64,23,7,8,34,1,543,36,34,34,24,23,]
{23,23,7,8,1,23,64,543,36,34,34,24,34,]
{23,23,7,8,1,]
{1,23,7,8,23,]
{23,7,8,23,]
```

```
{23,7,8,23,]
{23,7,8,]
{7,8,23,]
{64,543,36,34,34,24,34,]
{34,34,24,34,543,36,64,]
{34,34,24,]
{24,34,34,]
{34,34,]
{34,34,]
{543,36,64,]
{36,64,543,]
{1,7,8,23,23,23,24,34,34,34,36,64,543,}
 */
```

FIND CHARACTER INDEX STRING

```
import java.util.Scanner;
public class string {

    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.println("enter string");
        String element=sc.next();
        System.out.println("enter character ");
       char find=sc.next().charAt(0);
    for(int i=0;i<element.length();i++){
        if(element.charAt(i)==find){
            System.out.println("index  "+i);

            }

        }

    }
}
/*
```

OUTPUT

```
enter string
akhtar
enter character
t
index  3
 */
```

JUST ENJOYING

```
import java.util.Scanner;
import java.util.*;
class Data{
    static Scanner sc=new Scanner(System.in);
    private String[] phone_number=new String[10];
    private String email[]=new String[10];
    private  String[] Name=new String[10];
    private String adress[]=new String[10];
    private String Father_name[]=new String[10];

    Data(){
        System.out.println("Use (-) instead of space b/w two words ");
        phone_number[0]="0378676889";
        phone_number[1]="03333435363";
        phone_number[2]="03485448301";
        email[0]="dagaiAkhtar@gmail.com";
        email[1]="akhtardagai@gmail.com";
        email[2]="maazdagai@gmail.com";
        Name[0]="mudassir-ahmad";
        Name[1]="Akhtar-ali";
        Name[2]="maaz";
        adress[0]="dagai-buner";
        adress[1]="totali-dagai-buner";
        adress[2]="nilore-islamabad";
        Father_name[0]="bashir-ahmad";
        Father_name[1]="bashir-ahmad";
        Father_name[2]="muhammad-tayyab";

        System.out.println("Add new data press <1> ,not press any number ,");
        int choice=sc.nextInt();
        if(choice==1){
        System.out.println("enter size ");
        int size=sc.nextInt();
            for(int i=3;i<3+size;i++){
        System.out.println("enter phone number");
        phone_number[i]=sc.next();
        System.out.println("enter email");
        email[i]=sc.next();
        System.out.println("enter Name");
        Name[i]=sc.next();
        System.out.println("enter adress");
        adress[i]=sc.next();
        System.out.println("enter Father name");
```

```
        Father_name[i]=sc.next();
        System.out.println(" \n");

                }

            }else{

                }
        }

    void check(int i){
switch(i){
    case 1:
             System.out.println("**Enter Number**");

            String checknumber=sc.next();
            System.out.println("All data");
            for(int j=0;j<phone_number.length;j++){
               if(checknumber.equalsIgnoreCase(phone_number[j])){
               System.out.println("Phone number ::"+phone_number[j]);
               System.out.println("Name::"+Name[j]);
               System.out.println("Email::"+email[j]);
            System.out.println("adress::"+adress[j]);
            System.out.println("Father name::"+Father_name[j]);

               }
               else{
        System.out.println("sorry your number is not found in data base");
        }
    }
break;

case 2:
System.out.println("**Enter Name**");

            String checkName=sc.next();
            System.out.println("All data");
            for(int k=0;k<Name.length;k++){
               if(checkName.equalsIgnoreCase(Name[k])){
               System.out.println("Phone number ::"+phone_number[k]);
               System.out.println("Name::"+Name[k]);
               System.out.println("Email::"+email[k]);
            System.out.println("adress::"+adress[k]);
            System.out.println("Father name::"+Father_name[k]);

               }
               else{
        System.out.println("sorry your number is not found in data base");
```

```
    break;  }
    }
    break;
    case 3:
    System.out.println("**Enter  email**");

            String checkemail=sc.next();
            System.out.println("All data");
            for(int j=0;j<email.length;j++){
               if(checkemail.equalsIgnoreCase(email[j])){
               System.out.println("Phone number ::"+phone_number[j]);
               System.out.println("Name::"+Name[j]);
               System.out.println("Email::"+email[j]);
            System.out.println("adress::"+adress[j]);
            System.out.println("Father name::"+Father_name[j]);

               }
               else{
        System.out.println("sorry your email is not found in data base");
    break;  }
    }
break;
case 4:
    System.out.println("**Enter  adress**");

            String checkadress=sc.next();
            System.out.println("All data");
            for(int j=0;j<adress.length;j++){
               if(checkadress.equalsIgnoreCase(adress[j])){
               System.out.println("Phone number ::"+phone_number[j]);
               System.out.println("Name::"+Name[j]);
               System.out.println("Email::"+email[j]);
            System.out.println("adress::"+adress[j]);
            System.out.println("Father name::"+Father_name[j]);

               }
               else{
        System.out.println("sorry your adress is not found in data base");
    break;  }
    }


}
    }
    }
class showdata extends Data {
  void Show(){
```

```
        System.out.println("check by \n <phone number press 1>\n <Name press
2>\n <email press 3>\n <adress press 4>\n <Father name press 5>");
        int choices=sc.nextInt();
            check(choices);

        }

    public static void main(String[] args) {
        showdata obj=new showdata();
        obj.Show();

    }
}
```